

(a) the same arithmetic, three names

neutral

pen=1 + cup, cup=2 + 3; pen=?

the name agrees

pen=1 + five, five=2 + 3; pen=?

five really is 5

the name lies

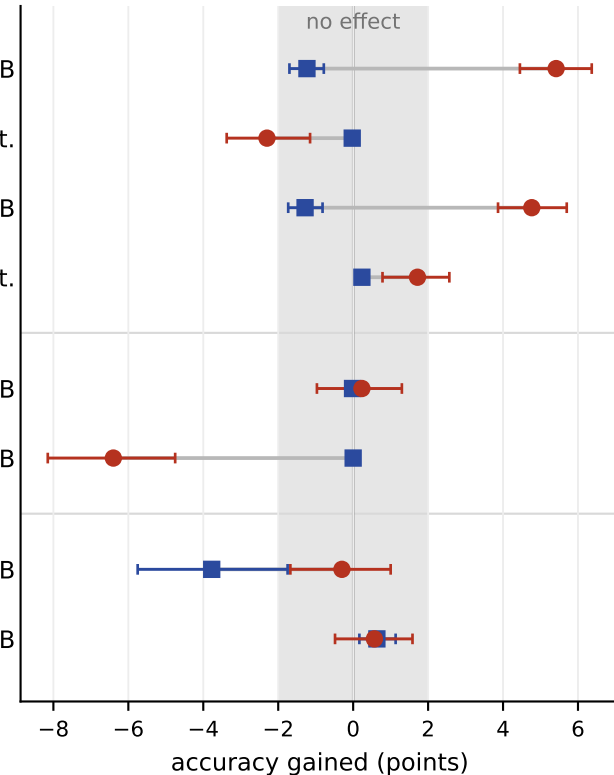
pen=1 + two, two=2 + 3; pen=?

two is really 5

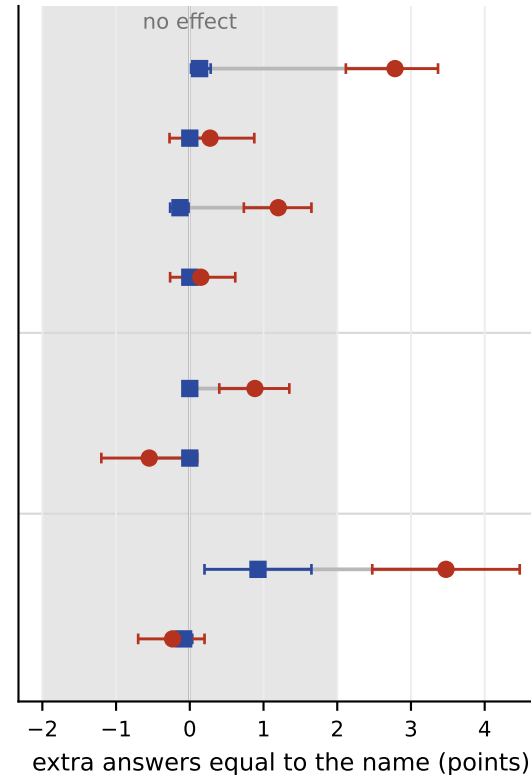
The answer is 6 in all three.

The model either writes the steps in full, ending "pen=1 + 5, pen=6", or answers "pen=6" directly.

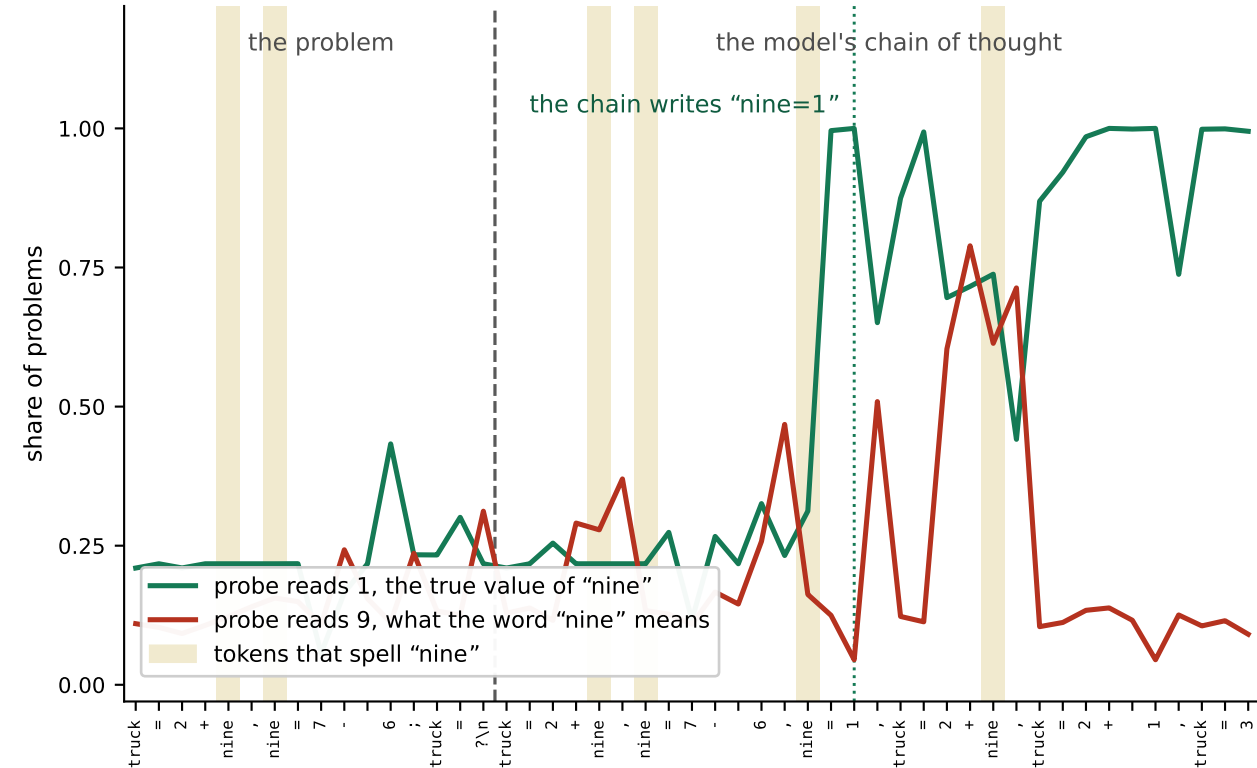
(b) a name that agrees: is the answer more often right?



a name that lies: is it given as the answer?



(c) what the model holds, token by token



● no chain of thought ■ with chain of thought ■ within 2 points of no effect